

2021*Time : 3 hours**Full Marks : 70*

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section–A**(Compulsory)**

1. Pick up the correct alternative for each of
the following questions : 10×2=20

(a) Which of the following is the correct
syntax of including a user defined header
file in C++ ?

- (i) `#include [userdefined]`
- (ii) `#include "userdefined"`
- (iii) `#include <userdefined.h>`
- (iv) `#include <userdefined>`

(b) Which of the following is used for comments in C++ ?

- (i) `/*comment*/`
- (ii) `//comments*/`
- (iii) `//comment`
- ☒ (iv) Both (i) & (iii)

(c) Which of the following user defined header file extension used in C++ ?

- (i) `hg`
- ☒ (ii) `cpp`
- (iii) `h`
- (iv) `hf`

(d) Which of the following is not a type of constructor in C++ ?

- (i) Default
- (ii) Parameterized
- (iii) Copy
- ✓ (iv) Friend

(e) Which of the following approach is used by C++ ?

- (i) Left-right
- (ii) Right-left
- ✓ (iii) Bottom up
- (iv) Top down

(f) Which is more effective while calling the C++ functions ?

- (i) Call by object
- (ii) Call by pointer

(iii) Call by value

☒ (iv) Call by reference

(g) Which of the following is used to terminate the function declaration in C++ ?

☒ (i) ;

(ii)]

(iii))

(iv) :

(h) Which keyword is used to define the macros in C++ ?

(i) #macro

☒ (ii) #define

(iii) macro

(iv) define

(i) The C++ code which causes abnormal termination/behaviour of a program should be written under ____ block.

(i) catch

(ii) throw

☒ (iii) try

(iv) finally

(j) What is meant by polymorphism ?

(i) Class having only single form

(ii) Class having four forms

☒ (iii) Class having many forms

(iv) Class having two forms

Section-B

Answer any **four** questions :

4×5=20

☒ 2/ How is C++ different from C language ? What are the object oriented features ?

3. What is the difference between structure and union ? Give example.
4. How are object passed as function arguments ?
What is difference between pass by value and pass by reference ?
5. Give suitable program to describe the idea behind friend function.
6. Write a program to illustrate function overloading.
7. What is token ? Explain the different types of tokens.
8. Differentiate between OOP and POP.

Section-C

Answer any **two** questions of the following :

$$2 \times 15 = 30$$

9. Explain the concept of inheritance ? What are the use of inheritance ? What are the different types available ?

10. What is a operator ? Explain the different operators available.

✓ 11. Write C++ program to print the following pattern :

(a) *
 * *

 * * *

 * * * *

```
#include<iostream.h>
void main()
{
    int i,j;
    for(i=1;i<=4;i++)
    {
        for(j=1;j<=i;j++)
        {
            cout<<"* ";
        }
        cout<<endl;
    }
}
```

(b) 1
 1 2
 1 2 3
 1 2 3 4

```
#include<iostream.h>
void main()
{
    int i,j;
    for(i=1;i<=4;i++)
    {
        for(j=1;j<=i;j++)
        {
            cout<<j;
        }
        cout<<endl;
    }
}
```

✓ 12. What is a loop ? What are the different types of loop available ? Explain with example.